

CRYST Branch Book

Diffraction, Lattice Periodicity, and Certified Crystallographic Structure

Elements of Nested Fibrational Cosmology
Prospective Derived Branch

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1. Legitimacy Analysis and Branch Constitution

Before constituting the branch, this section records the systematic legitimacy analysis required by Book V's Certified Branch Legitimacy Theorem ([thm:V.8.1](#)) and the UBLT ([thm:V.7.4](#)). The analysis is the basis for the verdict in Prop. 1.3; the branch is constituted only if all five UBLT conditions are satisfied.

1.1. The Question.

REMARK 1.1 ([R]). **Crystallography and NFC: the central question.** Standard crystallography operates on three imported primitives that the NFC framework forbids: (i) Euclidean three-dimensional geometry (a continuous background space); (ii) Fourier analysis (the reciprocal lattice and structure factors); (iii) atomic species (Na, Cl, C, ... as chemical primitives carrying pre-given scattering properties). The central question of this branch is whether crystallographic structure can be realized as a lawful descendant of **U** from purely finite observational primitives, without any of these three imports.

Specifically: starting from finite probe-response pairs (a beam probe, an intensity response), a finite admissible contrast discipline on the response data, and the observational equivalence imposed by that discipline, can one recover certified lattice periodicity, space group symmetry, and diffraction peak structure? Each of these notions, if it appears at all, must emerge as a quotient-visible, source-descended object within the declared regime.

The answer, given in Prop. 1.3, is: the crystallography program constitutes a lawful branch candidate. All five UBLT conditions can be satisfied at the declared pre-closure scope.

1.2. Five UBLT Conditions: Pre-Analysis.

REMARK 1.2 ([R]). **Pre-analysis of Book V UBLT conditions.** The five UBLT conditions (Book V, [thm:V.7.4](#)) and their crystallographic realizations:

(1) **Declared branch candidate.** The crystallography branch candidate $\mathcal{B}_{\text{Cryst}}$ is the program targeting certified diffraction-peak structure, lattice periodicity, and space group symmetry on the declared regime $\mathcal{R}_{\text{Cryst}}$. The five mandatory components (realization functor F_{Cryst} , burden package $\text{Cert}_{\text{Cryst}}$, terminal predicate τ_{Cryst} , failure frontier FF_{Cryst} , and observable family $\mathcal{O}_{\text{Cryst}}$) are declared below. *Condition (1): satisfied.*

(2) **Observable family visibility.** The primary diffraction observable is the Bragg intensity $I(hkl) = |F_{\text{Cryst}}(hkl)|^2$. This is the squared modulus of the structure factor: it is a nonnegative real number measured at each reciprocal lattice point (hkl) and is directly observable under admissible diffraction probes. The intensity is quotient-visible: two crystal configurations indistinguishable by all admissible diffraction probes have the same intensity profile. The Bragg gap $B_{\text{Cryst}} > 0$ (positive separation of Bragg peaks from the diffuse background) is also quotient-visible: it is the minimum intensity contrast between the peak maximum and the inter-peak floor, measurable from the certified intensity profile. *Condition (2): satisfied.*

(3) Source-descended comparison. The diffraction probe-response structure descends from **U** via the same route as the SPEC branch: probe episodes are admissible processes on the certified observable class; their responses are certified observable outcomes; the comparison discipline is the observational quotient of Book I applied to intensity profiles. The lattice periodicity structure descends from **U** via the Book III transport law applied to the admissible equivalence under lattice translations. *Condition (3): satisfied.*

(4) No hidden supplement. Three potential hidden imports are identified and addressed:

- (a) *Euclidean geometry*: not imported as primitive. The three-dimensional spatial structure of the crystal must emerge as the effective encoding of certified discrete lattice data, licensed by a Book VI bridge (O-CRYST.EWALD).
- (b) *Fourier analysis*: the reciprocal lattice and structure factor notation are licensed only after the discrete diffraction peaks are certified and the Book VI asymptotic coherence bridge (O-CRYST.EWALD) is established.
- (c) *Atomic species*: atoms are not primitives. A *scatterer* in this branch is characterized solely by its certified probe-response behavior (scattering length or electron density contribution), not by its chemical identity. Chemical identity, if it appears at all, must emerge from the observable scattering data.

Condition (4): satisfied (pending O-CRYST.EWALD for the Fourier/Euclidean bridge).

(5) No path multiplication. The crystallography branch’s terminal predicate (certified Bragg peak structure with positive gap and space group descent) is not the terminal predicate of any existing branch:

- YM targets a spectral mass gap for the covariant Laplacian on the gauge sector — distinct from a diffraction gap;
- SPEC targets probe-response resonance structure — adjacent but not identical (SPEC does not target periodicity or space group structure);
- BIO targets replication-heredity structure — distinct;
- LING targets contrast-compositional structure — distinct;
- SCC targets structural counterfactual capacity — distinct.

The crystallography terminal predicate is distinct from all existing named branches.

Condition (5): satisfied.

1.3. Legitimacy Proposition.

PROPOSITION 1.3 ([U]). (Crystallography Branch Is Lawful.) *The crystallography branch candidate $\mathcal{B}_{\text{Cryst}}$ satisfies all five conditions of the Certified Branch Legitimacy Theorem (Book V, **thm:V.8.1**). It is a lawful NFC branch candidate.*

PROOF. Each of the five UBLT conditions is addressed in Rem. 1.2: (1) all five mandatory components are declared in §2–8; (2) the Bragg intensity $I(hkl)$ is quotient-visible and the Bragg gap $B_{\text{Cryst}} > 0$ is branch-visible; (3) source descent follows the SPEC branch template via Book I observational quotient and Book III transport law; (4) no Euclidean geometry, Fourier analysis, or atomic species is imported as primitive — each is either earned by a Book VI bridge or dismissed by the no-smuggling standing

rules; (5) the terminal predicate is distinct from all existing branches by the terminal-predicate distinctness argument. By Book V $\mathbf{thm:V.8.1}$, $\mathcal{B}_{\text{Cryst}}$ is a lawful branch candidate. $\square$ $\square$

PROPOSITION 1.4 ([U]). (Full Distinctness.) *The crystallography branch is genuinely distinct from all existing named branches. Its terminal predicate — certified Bragg peak structure with positive gap $B_{\text{Cryst}} > 0$ and source-descended space group classification — is not the terminal predicate of YM, NS, GR, SCC, RH, SM, SPEC, BIO, or LING.*

PROOF. By the pairwise comparison in Rem. 1.2(5): the Bragg gap is not the YM mass gap (different carrier and probe family); not the SPEC spectral margin (SPEC targets resonance structure, not lattice periodicity); not the BIO fidelity margin (different observable class); not the LING contrast margin (different branch regime). No other branch targets space group symmetry as its terminal predicate. $\square$ $\square$

PROPOSITION 1.5 ([U]). (Crystallography Anti-Smuggling Gate.) *No branch claim may import Euclidean geometry, Fourier analysis, or atomic species as branch primitives. These must be either (a) derived as effective encodings of certified discrete observable structure via Book VI bridges, or (b) dismissed by the no-smuggling standing rules.*

PROOF. By Book I Standing Rule I.1.3 (no-smuggling at the primitive level) and the branch-level extension (Book V UBLT condition (4)): any datum outside the certified source image $F_{\text{Cryst}}(\mathbf{U})$ is a silent import and violates the intake law. $\square$ $\square$

2. Purpose and Canonical Seed

This branch asks whether crystallographic structure — lattice periodicity, space group symmetry, diffraction peak separation, and structure determination — can be realized as a lawful descendant of the NFC source architecture. The aim is not to import the standard language of Euclidean lattices, Fourier transforms, Miller indices, or atomic coordinates as primitives, but to determine when such language is licensed as an effective encoding of certified discrete observable structure within a declared branch regime. This posture is mandated by Book I’s no-smuggling stance, Book V’s branch-legitimacy law, Book VI’s continuum-interface discipline, and Book VII’s scoped-certificate law.

The branch has three structural precedents in the existing corpus. First, the SPEC branch already treats probe-response pairs and transition ledgers as the building blocks of a certified observable family — the *diffraction-probe template*: a beam probe acts on a crystal configuration and produces an intensity response, which is a SPEC-style probe-response pair. Second, the BIO branch already treats a replication-maintaining core I_X and boundary-forcing exchange as the route to a certified interface — applicable here because crystal grain boundaries and unit cell boundaries share the same boundary-capacity structure. Third, the SCC branch’s minimal structural carrier (MSC) machinery provides the anti-inflation gate for the space group identification: among all symmetry objects compatible with a given diffraction pattern, MSC selects

the unique minimal faithful carrier — which, at the declared regime, is one of the 230 crystallographic space groups.

The initial endpoint is deliberately modest: certified Bragg peak structure on a declared crystallographic regime, with positive peak separation from the diffuse background (the Bragg gap), and source-descended translation equivalence classes (the lattice structure). Full space group identification, structure determination, and the resolution of the phase problem are deferred as named open obligations.

REMARK 2.1 ([R]). Strategic path. The shortest canonical route is: probe-response certification (diffraction as admissible probe) $\Rightarrow$ Bragg gap (positive peak separation) $\Rightarrow$ lattice certification (translation equivalence classes) $\Rightarrow$ space group descent (MSC-selected symmetry carrier) $\Rightarrow$ endpoint. The phase problem and structure determination are downstream open obligations, expected to require Book VI bridges and possibly new probe-family extensions beyond intensity-only measurement.

3. Imported Spine Structure

Imported spine results.

- (1) *Book I*: observational quotient discipline; no Euclidean geometry, Fourier basis, atomic species, or crystallographic convention may be assumed. Crystal content must be quotient-visible and admissibly testable.
- (2) *Book II*: boundary capacity law. Crystal grain boundaries and unit cell boundaries are interface structures whose information capacity is logarithmically bounded. This governs the admissible probe resolution of crystal interfaces.
- (3) *Book III*: unified coercive-transport inequality. Stability costs distribute across the crystal lattice without hidden channel. Lattice phonons and thermal diffuse scattering are accounted for as defect contributions to the transport ledger.
- (4) *Book IV*: universal source $\mathbf{U}$ and source-descended comparison. All certified crystallographic objects factor through $\mathbf{U}$.
- (5) *Book V*: lawful branchhood; the constitutional law under which the crystallography branch is admitted (Prop. 1.3).
- (6) *Book VI*: continuum language licensed only as faithful shorthand for certified discrete structure within a declared interface regime. The Book VI gate controls: continuous angle/wavelength notation (Bragg's law as asymptotic shorthand), Fourier/reciprocal lattice notation (licensed only after O-CRYST.EWALD is discharged), and Euclidean $\mathbb{R}^3$ domain notation (licensed only after O-CRYST.DOM is discharged).
- (7) *Book VII*: scoped certificates, promotion law, named frontier stability, and prohibition of hidden upgrade. Claims about universal crystallographic laws, space group uniqueness, or structure determination from intensities alone are subject to this governance without exception.

Imported branch precedents.

- (1) *SPEC branch*: probe-response objects and transition ledgers — the *diffraction-probe template* for this branch. Diffraction is a SPEC-style probe episode; intensity data is a SPEC-style response observable.
- (2) *BIO branch*: boundary-forcing exchange and the replication-maintaining core — the *unit-cell boundary template*. A crystal unit cell’s interior/exterior separation follows the same boundary-forcing logic as BIO’s ∂I_X .
- (3) *SCC branch*: MSC machinery — the *space-group minimality template*. Among all certified symmetry objects compatible with a given intensity profile, MSC selects the unique minimal faithful carrier.
- (4) *YM branch*: gap structure — the Bragg gap $B_{\text{Cryst}} > 0$ is the fifth canonical instance of the positive-margin separation pattern (after YM mass gap, SPEC spectral margin, LING contrast margin, and BIO fidelity margin).

4. Branch-Specific Arena

4.1. Standing Rules.

STANDING RULE 4.1 ([D]). (No Smuggling.) No crystallographic object — Bravais lattice, space group, Miller index, structure factor, atomic coordinate, or unit cell — may be treated as primitive branch data merely because it is familiar in external crystallographic theory. Every branch-visible crystallographic object must be sourced through **U**, the declared descent machinery, and the crystallography observable family. This is the branch-local instance of Book I no-smuggling and Book V branch-legitimacy discipline.

STANDING RULE 4.2 ([D]). (No Euclidean Primitive.) Euclidean three-dimensional geometry is not a primitive of this branch. The ambient space of the crystal, if it appears at all, must emerge as the effective encoding of certified discrete lattice data via a Book VI domain bridge (O-CRYST.DOM). Until that bridge is established, all spatial language is forbidden as a branch primitive.

STANDING RULE 4.3 ([D]). (No Fourier Primitive.) Fourier analysis, the reciprocal lattice, structure factors as complex amplitudes, and Miller index notation as geometric primitive are not licensed until the Book VI Ewald bridge (O-CRYST.EWALD) is established. The only immediately certified diffraction observable is the Bragg intensity $I(hkl) = |F(hkl)|^2$, which is a real nonnegative probe-response datum — not the full complex structure factor.

STANDING RULE 4.4 ([D]). (No Atomic Primitive.) No atomic species, chemical element, or molecular formula may be treated as a primitive. A *scatterer* in this branch is characterized solely by its certified probe-response behavior within the declared admissible probe discipline. Chemical identity, if it appears at all, must emerge from the observable scattering data.

STANDING RULE 4.5 ([D]). (Quotient Visibility.) A crystallographic distinction between two configurations counts only when it is quotient-visible under admissible diffraction probe tests. Two crystal configurations whose intensity profiles agree on all

admissible diffraction probes belong to the same certified crystallographic equivalence class, regardless of their chemical or geometric description in external notation.

STANDING RULE 4.6 ([D]). (No Self-Promotion.) A finite certified diffraction result may not self-promote to a claim about all crystal structures, universal space group uniqueness, or complete structure determination from intensities alone. Such promotion requires an explicit transfer theorem. This is a direct Book VII governance consequence.

STANDING RULE 4.7 ([D]). (Frontier Visibility.) Named frontiers in space group identification, structure determination, the phase problem, domain certification, and Fourier legitimacy remain visible until discharged by theorem. Rephrasing does not discharge them.

STANDING RULE 4.8 ([D]). (Continuum Gate.) Continuous angle notation (2θ), wavelength notation (λ), Fourier transform notation for electron density, and Euclidean coordinate notation may appear only after the corresponding Book VI bridge theorem certifies that they faithfully encode certified discrete observable structure within the declared continuum interface regime.

4.2. Declared Target Regime.

DEFINITION 4.9 ([D]). (Crystallography Target Regime.) A *crystallography target regime* is a declared branch regime $\mathcal{R}_{\text{Cryst}}$ consisting of:

- (1) a certified observable class $\mathcal{C}_{\text{obs}}$ of physical configurations — the objects on which diffraction probe tests are performed (crystal specimens in the declared regime, not atoms or molecules);
- (2) a finite admissible diffraction probe discipline $\mathcal{P}_{\text{diff}}$, each element of which is a declared admissible process producing a quotient-visible intensity response;
- (3) a declared probe-energy window $\mathcal{W}_{\text{diff}}$ specifying the range of probe energies (wavelengths) within which diffraction claims have force; and
- (4) a declared comparison discipline governing when two intensity profiles $\{I(hkl)\}$ are equivalent for the purpose of crystallographic claims.

REMARK 4.10 ([R]). **What a configuration is.** A *physical configuration* in this branch is a certified observable arrangement of scatterers (not atoms or molecules — those are chemical primitives) on which admissible diffraction probes can be performed. Whether any such configuration qualifies as periodic, crystalline, or amorphous is determined by the diffraction kernel and lattice ledger, not by prior crystallographic classification.

5. Crystallography Observable Family

5.1. Three-Part Structure.

DEFINITION 5.1 ([D]). (Crystallography Observable Family.) A *crystallography observable family* on $\mathcal{R}_{\text{Cryst}}$ is a triple

$$\mathcal{O}_{\text{Cryst}} = (\Phi_{\text{Cryst}}, \mathcal{L}_{\text{Cryst}}, \mathcal{G}_{\text{Cryst}})$$

where:

- (1) Φ_{Cryst} is the *diffraction kernel*: the quotient-visible intensity profile induced on $\mathcal{C}_{\text{obs}}$ by the admissible diffraction probe discipline $\mathcal{P}_{\text{diff}}$;
- (2) $\mathcal{L}_{\text{Cryst}}$ is the *lattice ledger*: certified records of translation equivalence classes on the observable configuration, with full transport bookkeeping; and
- (3) $\mathcal{G}_{\text{Cryst}}$ is the *symmetry-descent object*: a branch-visible certified symmetry carrier governing the complete set of admissible symmetry operations on the diffraction-visible configuration.

5.2. Diffraction Kernel.

DEFINITION 5.2 ([D]). (Diffraction Kernel.) The *diffraction kernel* Φ_{Cryst} is the assignment

$$\Phi_{\text{Cryst}} : (X, P_{hkl}) \mapsto I(hkl) := |F(hkl)|^2$$

from a configuration X and admissible diffraction probe $P_{hkl} \in \mathcal{P}_{\text{diff}}$ to the certified intensity response. Two configurations belong to the same *diffraction equivalence class* $[X]_{\Phi}$ if and only if no admissible probe in $\mathcal{P}_{\text{diff}}$ distinguishes their intensity profiles.

REMARK 5.3 ([R]). **The phase is not part of the kernel.** The diffraction kernel records $I(hkl) = |F(hkl)|^2$ — the squared modulus of the structure factor. The phase $\phi(hkl) = \arg F(hkl)$ is not part of the intensity-observable kernel. Whether phase information is quotient-invisible (cannot be recovered from intensity data alone under any admissible probe discipline) or recoverable via richer probes is the content of the principal frontier O-CRYST.PHASE (§7.8). This is the most structurally interesting feature of the crystallography branch: the *phase problem* maps directly onto the NFC question of what is and is not quotient-visible under the declared admissible probe discipline.

DEFINITION 5.4 ([D]). (Bragg Gap.) The *Bragg gap* at a configuration X is

$$B_{\text{Cryst}}(X) := \min_{hkl \in \text{peak}(X)} [I(hkl) - I_{\text{bg}}(X)],$$

where $\text{peak}(X)$ is the certified set of Bragg peak indices and $I_{\text{bg}}(X)$ is the certified diffuse background intensity at those positions. The Bragg gap is positive when diffraction peaks are discretely separated from the background: $B_{\text{Cryst}}(X) > 0$.

REMARK 5.5 ([R]). **The Bragg gap as the fifth canonical gap instance.** $B_{\text{Cryst}}(X) > 0$ is the crystallography branch's instance of the canonical positive-margin separation structure that appears across the NFC corpus: the YM mass gap ($m^2 > 0$), the SPEC spectral margin ($\lambda_1 > 0$), the LING contrast margin ($\Delta_{\text{Ling}} > 0$), and the BIO replication fidelity margin ($F_{\text{Bio}} > 0$). In each case the structural claim is a positive observable separation between distinct certified equivalence classes. Here: Bragg peaks are discretely separated from the diffuse scattering background in a certified intensity-observable sense.

5.3. Lattice Ledger.

DEFINITION 5.6 ([D]). (Lattice Ledger.) The *lattice ledger* $\mathcal{L}_{\text{Cryst}}$ is a certified bookkeeping object recording, for a configuration X :

- (1) the certified set of translation equivalence classes $\{[x]_{\sim_t}\}$ — the quotient of the configuration's admissible sites under the certified translation probe discipline;
- (2) the certified transport cost and defect contribution of each admissible translation, governed by Book III; and
- (3) the Book II boundary capacity bookkeeping for the unit cell boundary ∂U_X (the interface between one translation equivalence class and its neighbors).

The lattice ledger is the crystallographic analogue of the compositional ledger (LING branch) and the metabolic ledger (BIO branch).

REMARK 5.7 ([R]). Lattice as quotient structure. The Bravais lattice, in standard crystallographic terms, is the group of translation vectors that map the crystal onto itself. In NFC terms, it is the quotient of the configuration's admissible sites under the certified translation probe equivalence: two sites x, x' are translation-equivalent if no admissible diffraction probe distinguishes them when one is mapped to the other by an admissible process. The lattice is derived, not assumed. Whether it is one of the 14 Bravais lattice types is a derived theorem requiring Book VI geometric legitimacy, not a branch assumption.

5.4. Symmetry-Descent Object.

DEFINITION 5.8 ([D]). (Symmetry-Descent Object.) A *symmetry-descent object* $\mathcal{G}_{\text{Cryst}}$ is a branch-visible certified symmetry carrier that:

- (1) is sourced through **U**;
- (2) is selected by MSC/minimal-support discipline as the unique minimal faithful carrier for the diffraction-visible symmetry predicate on the declared regime;
- (3) produces only quotient-visible distinctions among configurations; and
- (4) passes the Book V five-condition branch-legitimacy test.

$\mathcal{G}_{\text{Cryst}}$ is not a space group declared by external crystallographic convention. It is a derived certified carrier. If, under the declared regime and admissible probe discipline, MSC selects a carrier matching one of the 230 crystallographic space groups, that match is a derived theorem — the content of O-CRYST.SG — not a branch assumption.

REMARK 5.9 ([R]). The 230 space groups as a prediction, not an assumption. Standard crystallography enumerates 230 crystallographic space groups as the complete set of symmetry groups compatible with three-dimensional Euclidean lattice periodicity. In the NFC framework, this enumeration must be derived: the claim is that MSC applied to the diffraction-visible symmetry predicate on the declared Euclidean domain (after O-CRYST.DOM and O-CRYST.EWALD are discharged) selects carriers that, up to branch-visible equivalence, coincide with one of these 230 groups. This is a precise, falsifiable prediction of the branch — its compact derivable content. If MSC selects more or fewer carriers than 230, the branch has discovered either a new symmetry class or an error in the existing enumeration.

6. Endpoint Datum and Branch Posture

DEFINITION 6.1 ([D]). (Crystallography Endpoint Datum.) The *crystallography endpoint datum* is the branch-visible package

$$E_{\text{Cryst}} = (\mathcal{C}_{\text{obs}}, \mathcal{O}_{\text{Cryst}}, B_{\text{Cryst}}, \mathcal{W}_{\text{diff}})$$

consisting of the certified observable class, the crystallography observable family, the Bragg gap, and the declared probe-energy window.

DEFINITION 6.2 ([D]). (Lawful Crystallography Descendant Claim.) A *lawful crystallography descendant claim* is a claim that the endpoint datum E_{Cryst} is sourced through $\mathbf{U}$ and supports a certified diffraction, periodicity, or symmetry statement on the declared regime.

PROPOSITION 6.3 ([C]). (Crystallography Branch Candidate.) [Conditional on the crystallography target regime, observable family, and endpoint datum all being sourced through $\mathbf{U}$, branch-visible, and admitting a Book V legitimacy witness.] *The crystallography program is a lawful branch candidate.*

PROOF. By Prop. 1.3 and the branch-legitimacy doctrine (Book V) applied to the declared crystallography endpoint datum. $\square$ $\square$

7. Core Theorem Shells and Named Frontiers

7.1. Probe Realization.

OPEN OBLIGATION 7.1 ([O]). (O-CRYST.PR: Probe Realization.) Show that the diffraction probe family $\mathcal{P}_{\text{diff}}$ is sourced through $\mathbf{U}$ and constitutes a lawful branch observable within the declared regime, with all probe declarations inherited from the Book I admissible-test discipline.

THEOREM 7.2 ([C]). (Probe Realization Theorem.) [Conditional on O-CRYST.PR.] *The diffraction kernel Φ_{Cryst} is a lawful branch observable descended from $\mathbf{U}$. The certified diffraction equivalence classes $\{[X]_{\Phi}\}$ are quotient-visible and admissibly testable.*

PROOF. By the branch-legitimacy doctrine (Book V) applied to the declared diffraction kernel, together with the Book I observational quotient discipline. Full proof is conditional on O-CRYST.PR. $\square$ $\square$

7.2. Bragg Gap.

OPEN OBLIGATION 7.3 ([O]). (O-CRYST.GAP: Bragg Gap.) Show that the Bragg gap $B_{\text{Cryst}}(X)$ is branch-computably positive for certified crystalline configurations within the declared regime, and that no Bragg peak collapse occurs without a declared admissible structure-change event.

THEOREM 7.4 ([C]). (Bragg Gap Theorem.) [Conditional on O-CRYST.PR and O-CRYST.GAP; the admissible diffraction probe family being non-trivially separating on the declared regime.] *For any certified crystalline configuration X , $B_{\text{Cryst}}(X) > 0$: Bragg peaks are discretely separated from the diffuse scattering background.*

PROOF. By the observational quotient discipline (Book I) and the declared admissible-test finiteness condition on $\mathcal{R}_{\text{Cryst}}$. The positivity of the Bragg gap reflects the separating power of the admissible diffraction probe discipline. Full proof is conditional on O-CRYST.GAP. $\square$ $\square$

7.3. Lattice Certification.

OPEN OBLIGATION 7.5 ([O]). (O-CRYST.LAT: Lattice Certification.) Show that the lattice ledger $\mathcal{L}_{\text{Cryst}}$ is additive, sourced through the Book III transport accounting, and that the translation equivalence classes are genuinely quotient-visible (no hidden spatial primitive is required).

THEOREM 7.6 ([C]). (Lattice Certification Theorem.) [Conditional on O-CRYST.PR and O-CRYST.LAT.] *The lattice ledger $\mathcal{L}_{\text{Cryst}}$ is additive over declared translation steps. Transport costs distribute without hidden channel; translation equivalence classes are quotient-visible.*

PROOF. By the Book III unified coercive-transport inequality applied to the translation operations, together with the no-smuggling standing rule (Sr. 4.2). Full proof is conditional on O-CRYST.LAT. $\square$ $\square$

7.4. No Hidden Diffraction Channel.

OPEN OBLIGATION 7.7 ([O]). (O-CRYST.NC: No Hidden Diffraction Channel.) Show that no diffraction pathway exists outside the declared admissible probe family: any discrepancy between two configurations sharing the same certified diffraction equivalence class would require an undeclared diffraction channel, ruled out by the exhaustive source-image condition.

THEOREM 7.8 ([C]). (No-Hidden-Diffraction-Channel Theorem.) [Conditional on O-CRYST.LAT and the exhaustive source-image condition (`def:exhaustive-source-image`, Book IV).] *Two configurations sharing the same certified diffraction equivalence class have the same Bragg intensity profile under all admissible probes. No undeclared diffraction channel can produce a branch-visible intensity distinction invisible to the declared intensity ledger.*

PROOF. Any discrepancy would require an undeclared diffraction channel. By the exhaustive source-image condition applied to $\mathbf{U}$ and the no-Fourier-primitive standing rule (Sr. 4.3), no such channel is admitted. Full proof is conditional on O-CRYST.NC. $\square$ $\square$

7.5. Space Group Descent.

OPEN OBLIGATION 7.9 ([O]). (O-CRYST.SG: Space Group Descent.) Show that the symmetry-descent object $\mathcal{G}_{\text{Cryst}}$ is sourced through $\mathbf{U}$, passes the Book V five-condition legitimacy test, and is selected by MSC/minimal-support discipline as the unique minimal faithful carrier for the diffraction-visible symmetry predicate. Under the declared Euclidean domain (after O-CRYST.DOM and O-CRYST.EWALD), show that $\mathcal{G}_{\text{Cryst}}$ coincides with one of the 230 crystallographic space groups.

THEOREM 7.10 ([C]). (Space Group Descent Theorem.) [Conditional on O-CRYST.PR, O-CRYST.NC, O-CRYST.SG, and the declared Euclidean domain (O-CRYST.DOM).] *The symmetry-descent object $\mathcal{G}_{\text{Cryst}}$ is a lawful branch-visible certified symmetry carrier. Its MSC-selected faithful subcarrier coincides with one of the 230 crystallographic space groups within the declared domain scope.*

PROOF. By the Book V branch-legitimacy doctrine applied to $\mathcal{G}_{\text{Cryst}}$, together with the no-smuggling and no-Euclidean-primitive standing rules. MSC selects the minimal faithful carrier by the Zorn/finite-stabilization argument (`thm:YM-MS`, YM Branch §5A, transplanted to the crystallographic symmetry-carrier family). The coincidence with one of the 230 space groups requires O-CRYST.DOM (for the domain) and O-CRYST.EWALD (for the geometric structure). Full proof is conditional on O-CRYST.SG. $\square$ $\square$

7.6. Domain Bridge.

OPEN OBLIGATION 7.11 ([O]). (O-CRYST.DOM: Euclidean Domain Bridge.) Show that the crystallography branch has a certified three-dimensional effective geometric carrier D_{Cryst} , descended from **U** via a Book VI domain bridge, satisfying $\dim D_{\text{Cryst}} = 3$ and matching the Euclidean continuum domain at observable/operator/no-loss quotient scope. This is the crystallographic analogue of the YM B2 domain bridge (`prop:ClaySpec-DOM`, YM Branch).

REMARK 7.12 ([R]). **Why the domain bridge is required.** The crystal's ambient three-dimensional space is not a primitive. It must emerge as the effective encoding of certified discrete lattice data via Book VI. The domain bridge proceeds exactly as in the YM B2 analysis: build a domain reconstruction algebra $A_{\text{Cryst,dom}}^\infty$ from finite certified data (diffraction observables, lattice translation records, boundary collar data), show that its character space contains a nontrivial three-dimensional carrier, verify Van Hove/collar exhaustion and topological non-load-bearing cleanup, and establish no-loss passage from the certified discrete lattice to the Euclidean continuum.

7.7. Ewald Bridge (Fourier Legitimacy).

OPEN OBLIGATION 7.13 ([O]). (O-CRYST.EWALD: Ewald/Fourier Bridge.) Show that reciprocal lattice notation, structure factor notation ($F(hkl)$ as a complex amplitude), and the Bragg equation ($2d \sin \theta = n\lambda$) are licensed as effective shorthands for certified discrete diffraction peak data within a declared Fourier-interface regime. This is the Book VI bridge controlling all continuous-angle, continuous-wavelength, and complex-amplitude notation in the crystallography branch.

REMARK 7.14 ([R]). **The Ewald bridge as Book VI instance.** The Ewald construction in standard crystallography geometrically relates the direct lattice, the reciprocal lattice, and the sphere of reflection in Euclidean space. In NFC terms, the Ewald bridge is a Book VI continuum interface theorem: it licenses the reciprocal lattice as the asymptotic limit of a certified sequence of finite diffraction peak records, and it licenses the structure factor $F(hkl)$ as the asymptotic Fourier coefficient of the certified scattering density. The missing ingredient for the Book VI bridge is the asymptotic

coherence of the finite diffraction data in the declared Fourier-interface regime — the crystallographic analogue of O-LING.REC-Bridge- ε and O-BIO.EVO-Bridge- ε .

7.8. The Phase Problem: Principal Frontier.

OPEN OBLIGATION 7.15 ([O]). (O-CRYST.PHASE: Phase Problem — Principal Frontier.) The *phase problem* is the principal frontier of the crystallography branch. Diffraction experiments observe $I(hkl) = |F(hkl)|^2$ (intensities) but not the phases $\phi(hkl) = \arg F(hkl)$ of the structure factors. The NFC formulation of the phase problem is:

- (1) *Phase invisibility question*: Is $\phi(hkl)$ quotient-invisible under the declared admissible diffraction probe discipline? That is, is the phase information fundamentally lost in the observational quotient, or is it recoverable?
- (2) *Phase recovery route*: If phase information is not quotient-invisible, identify a richer admissible probe family $\mathcal{P}'_{\text{diff}} \supset \mathcal{P}_{\text{diff}}$ that makes the phase quotient-visible.
- (3) *Structure determination*: Given either (a) certified phase invisibility (showing that no admissible intensity-only experiment can determine the structure) or (b) a certified phase recovery probe family, show that the certified intensity and phase data together uniquely determine the diffraction equivalence class of the configuration.

REMARK 7.16 ([R]). **Phase problem as an NFC structural question.** The phase problem maps with unusual precision onto the NFC framework:

Why the phase problem is the principal frontier. Standard crystallography has developed numerous methods for recovering phase information (Patterson methods, direct methods, isomorphous replacement, anomalous dispersion, molecular replacement). These are all methods for extending the admissible probe family beyond intensity-only measurement: they add prior information (molecular geometry, heavy atoms, multiple wavelengths). In NFC terms, each such method is a candidate for $\mathcal{P}'_{\text{diff}}$: a richer probe family that makes more crystallographic information quotient-visible.

Route (a): phase invisibility. If the phase is quotient-invisible under intensity-only measurement ($\mathcal{P}_{\text{diff}}$), then no intensity-only experiment can distinguish two configurations that are related by a Babinet-complement or twin operation. This is a certified structural fact about the admissible probe discipline — a no-hidden-channel result analogous to O-LING.NC and O-BIO.NC.

Route (b): phase recovery. If the phase is recoverable via a richer probe family $\mathcal{P}'_{\text{diff}}$ (e.g., adding anomalous dispersion or multiple isomorphous replacement), the branch must certify that $\mathcal{P}'_{\text{diff}}$ is lawful (sourced through **U**, no hidden supplementation), and that the phase information extracted is quotient-visible under that richer family.

The NFC contribution. The NFC framework makes the phase problem precise and structural: it is not merely a practical measurement challenge, but a question about the information capacity of the admissible probe discipline. The Book II boundary capacity law bounds the maximum information extractable from any finite probe

episode; this bounds how much phase information can ever be certified from a finite diffraction experiment.

The phase problem is therefore the crystallography branch’s counterpart to the NS Stage-3 Clay frontier: both are genuinely open mathematical questions that the NFC framework makes precise.

7.9. Crystallography Endpoint.

OPEN OBLIGATION 7.17 ([O]). (O-CRYST.END: Crystallography Endpoint.) Show that the endpoint datum E_{Cryst} is fully certified: probe realization, Bragg gap, lattice certification, no-hidden-diffraction-channel, and space group descent all discharged. The endpoint is reached when the crystallography branch satisfies the shared-endgame schema of Book VII.

THEOREM 7.18 ([C]). (Crystallography Endpoint Theorem.) [Conditional on O-CRYST.PR, O-CRYST.GAP, O-CRYST.LAT, O-CRYST.NC, O-CRYST.SG, O-CRYST.DOM, O-CRYST.EWALD.] *The endpoint datum*

$$E_{\text{Cryst}} = (\mathcal{C}_{\text{obs}}, \mathcal{O}_{\text{Cryst}}, B_{\text{Cryst}}, \mathcal{W}_{\text{diff}})$$

is a certified lawful crystallography endpoint: certified diffraction kernel, additive lattice ledger, positive Bragg gap, no-hidden-diffraction-channel control, and space group descent sourced through $\mathbf{U}$ and MSC-selected at the declared Euclidean domain scope.

PROOF. Combine Thms. 7.2, 7.4, 7.6, 7.8, and 7.10. Apply the shared-endgame schema (Book VII, Prop. 6.11): part (i) is the source-controlled diffraction kernel descended from $\mathbf{U}$; part (ii) is the lattice ledger; part (iii) is the no-hidden-channel theorem together with the Bragg gap. Full proof is conditional on O-CRYST.PR through O-CRYST.EWALD. $\square$ $\square$

8. Closure Ledger

REMARK 8.1 ([R]). **Obligation and frontier summary.**

Label	Status	Description
O-CRYST.PR	[O]	Probe Realization: Φ_{Cryst} sourced through U
O-CRYST.GAP	[O]	Bragg Gap: $B_{\text{Cryst}} > 0$ for crystalline configurations
O-CRYST.LAT	[O]	Lattice Certification: additive ledger, no hidden spatial primitive
O-CRYST.NC	[O]	No Hidden Diffraction Channel: exhaustive source-image control
O-CRYST.SG	[O]	Space Group Descent: MSC-selected symmetry carrier; 230 prediction
O-CRYST.DOM	[O]	Euclidean Domain Bridge: $D_{\text{Cryst}} \simeq \mathbb{R}^3$ at obs/op scope
O-CRYST.EWALD	[O]	Ewald/Fourier Bridge: reciprocal lattice and structure factor notation
O-CRYST.PHASE	[O]	Phase Problem (principal frontier): phase visibility or recovery route
O-CRYST.END	[O]	Endpoint: full certification of E_{Cryst}

Discharge order. The natural discharge order is: PR $\rightarrow$ GAP $\rightarrow$ LAT $\rightarrow$ NC $\rightarrow$ DOM $\rightarrow$ EWALD $\rightarrow$ SG $\rightarrow$ END. The phase problem (O-CRYST.PHASE) is a named frontier orthogonal to the main chain: it does not gate the endpoint but represents the deepest structural question of the branch. END can be reached without resolving O-CRYST.PHASE: the diffraction-periodicity-symmetry endpoint is honest even without full structure determination.

Two-status reading. The crystallography branch admits two natural closure readings:

- (a) *Diffraction-periodicity-symmetry closure* (PR–END discharged, PHASE open): certified Bragg peak structure, additive lattice ledger, Bragg gap positive, and MSC-selected space group carrier — the minimal honest crystallographic endpoint.
- (b) *Structure determination closure* (PR–PHASE discharged): adds either certified phase invisibility (no intensity-only experiment determines the structure) or a certified richer probe family recovering phase information — allowing the certified intensity and phase data together to determine the diffraction equivalence class uniquely.

Posture (a) is the correct near-term target. Posture (b) is the genuine long-term ambition.

Cross-branch relations.

- (1) O-CRYST.PR imports the SPEC probe-realization template but has no pre-requisite obligations in the SPEC branch.
- (2) O-CRYST.SG imports the SCC MSC machinery but requires only O-SCC.MCS [C] (conditionally discharged, Session XLII), not the full SCC CERT-CLOSE chain.

- (3) O-CRYST.DOM imports the YM B2 domain bridge pattern (`prop:ClaySpec-DOM`); no YM obligation is a prerequisite, only the structural template.
- (4) O-CRYST.EWALD is analogous to O-LING.REC-Bridge- ε (asymptotic coherence condition); both are Book VI Fourier/continuum bridges.
- (5) O-CRYST.PHASE has no analogue in existing branches — it is the first branch in the NFC corpus where the principal frontier is an *observational information question* (what can be certified from intensity-only probe data?) rather than a structural construction question.

9. Branch Status Proposition

PROPOSITION 9.1 ([C]). (Crystallography Branch Status.) *The crystallography branch currently satisfies:*

- (1) **Lawful branch:** *confirmed (Prop. 1.3). All five Book V UBLT conditions satisfied.*
- (2) **Full distinctness:** *confirmed (Prop. 1.4). The Bragg gap and space group identification are not the terminal predicate of any existing branch.*
- (3) **Anti-smuggling gate:** *confirmed (Prop. 1.5). No Euclidean geometry, Fourier analysis, or atomic species may be imported as primitive.*
- (4) **Observable family:** *the three-part family $\mathcal{O}_{\text{Cryst}} = (\Phi_{\text{Cryst}}, \mathcal{L}_{\text{Cryst}}, \mathcal{G}_{\text{Cryst}})$ is architecturally constituted. All theorem shells carry named open obligations.*
- (5) **Status:** CERT-PROJ. Architecturally constituted, pre-closure. Nine named open obligations.
- (6) **Strongest admissible label:** *lawful branch candidate with certified diffraction probe-response structure (pending O-CRYST.PR), positive Bragg gap (pending O-CRYST.GAP), and source-descended space group carrier (pending O-CRYST.SG and O-CRYST.DOM).*
- (7) **Principal frontier:** *O-CRYST.PHASE (the phase problem) is the deepest structural question — the first NFC branch frontier that is a question about the information capacity of the admissible probe discipline rather than a structural construction question.*
- (8) **Cross-branch connections:** *SPEC branch (diffraction as probe-response), BIO branch (unit cell boundary as boundary-forcing exchange), SCC branch (space group as MSC-selected carrier), YM branch (Bragg gap as fifth canonical gap instance, B2 domain bridge as DOM template).*

SCHOLIUM 9.2 ([R]). The crystallography branch is frontier-honest. It does not import Bragg’s law, the reciprocal lattice, or the 230 space groups as assumptions. It claims only that certified diffraction probe-response structure, positive peak separation, and source-descended periodicity and symmetry can be realized from finite observational primitives under the NFC framework. The phase problem is named as the principal frontier because it is the most structurally novel question the branch poses: whether the information-theoretic barrier of intensity-only measurement is a fundamental quotient-invisibility result or a recoverable limitation. That question is precise, honest, and falsifiable.